

FIT1047 Assignment 1 – Part 2

Quah Xuanyu (35624914)

Question Truth Table:

X1	X2	X3	X4	Z1	Z2
0	0	0	0	1	0
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	0	1
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	0	1	1	0
1	0	1	0	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	0	1	1	0
1	1	1	0	1	1
1	1	1	1	0	1

Task 2.1: Boolean Algebra Expressions

1. Sum Of Products (SOP)

Table used for **Z1**

No.	X1	X2	X3	X4	Z1	Result in Boolean Expression
<u>1</u>	0	0	0	0	1	$\overline{X1} \overline{X2} \overline{X3} \overline{X4}$
<u>2</u>	0	0	0	1	0	–
<u>3</u>	0	0	1	0	1	$\overline{X1} \overline{X2} X3 \overline{X4}$
<u>4</u>	0	0	1	1	0	–
<u>5</u>	0	1	0	0	0	–
<u>6</u>	0	1	0	1	1	$\overline{X1} X2 \overline{X3} X4$
<u>7</u>	0	1	1	0	1	$\overline{X1} X2 X3 \overline{X4}$
<u>8</u>	0	1	1	1	1	$\overline{X1} X2 X3 X4$
<u>9</u>	1	0	0	0	0	–
<u>10</u>	1	0	0	1	1	$X1 \overline{X2} \overline{X3} X4$
<u>11</u>	1	0	1	0	1	$X1 \overline{X2} X3 \overline{X4}$
<u>12</u>	1	0	1	1	1	$X1 \overline{X2} X3 X4$
<u>13</u>	1	1	0	0	0	–
<u>14</u>	1	1	0	1	1	$X1 X2 \overline{X3} X4$
<u>15</u>	1	1	1	0	1	$X1 X2 X3 \overline{X4}$
<u>16</u>	1	1	1	1	0	–

Boolean Formurla for **Z1**:

$$\begin{aligned}
 Z1 = & \overline{X1} \overline{X2} \overline{X3} \overline{X4} + \overline{X1} \overline{X2} X3 \overline{X4} + \overline{X1} X2 \overline{X3} X4 + \overline{X1} X2 X3 \overline{X4} + \overline{X1} X2 X3 X4 \\
 & + X1 \overline{X2} \overline{X3} X4 + X1 \overline{X2} X3 \overline{X4} + X1 \overline{X2} X3 X4 + X1 X2 \overline{X3} X4 \\
 & + X1 X2 X3 \overline{X4}
 \end{aligned}$$

Table used for **Z2**

No.	X1	X2	X3	X4	Z2	Result in Boolean Expression
<u>1</u>	0	0	0	0	0	-
<u>2</u>	0	0	0	1	1	$\overline{X1} \overline{X2} \overline{X3} X4$
<u>3</u>	0	0	1	0	0	-
<u>4</u>	0	0	1	1	1	$\overline{X1} \overline{X2} X3 X4$
<u>5</u>	0	1	0	0	1	$\overline{X1} X2 \overline{X3} \overline{X4}$
<u>6</u>	0	1	0	1	0	-
<u>7</u>	0	1	1	0	1	$\overline{X1} X2 X3 \overline{X4}$
<u>8</u>	0	1	1	1	0	-
<u>9</u>	1	0	0	0	1	$X1 \overline{X2} \overline{X3} \overline{X4}$
<u>10</u>	1	0	0	1	0	-
<u>11</u>	1	0	1	0	1	$X1 \overline{X2} X3 \overline{X4}$
<u>12</u>	1	0	1	1	1	$X1 \overline{X2} X3 X4$
<u>13</u>	1	1	0	0	0	-
<u>14</u>	1	1	0	1	0	-
<u>15</u>	1	1	1	0	1	$X1 X2 X3 \overline{X4}$
<u>16</u>	1	1	1	1	1	$X1 X2 X3 X4$

Boolean Formula for Z2:

$$Z2 = \overline{X1} \overline{X2} \overline{X3} X4 + \overline{X1} \overline{X2} X3 X4 + \overline{X1} X2 \overline{X3} \overline{X4} + \overline{X1} X2 X3 \overline{X4} + X1 \overline{X2} \overline{X3} \overline{X4} + X1 \overline{X2} X3 \overline{X4} + X1 \overline{X2} X3 X4 + X1 X2 X3 \overline{X4} + X1 X2 X3 X4$$

Sum of Product (SOP) Explanation:

- Each of the rows that results in the output of 1, is converted into boolean **minterm**. For each of the input that is 0, the variable is complemented by putting it through a NOT gate, while input with 1 remain uncomplemented.
- The four variables are then combined into an AND gate.
- Once all the rows where the output is 1 has been converted to a minterm, an OR gate is used to combine all the outputs of the AND gates.
- For example, row 6 with input 0101 for Z1, X1 and X3 will be complemented, while X2 and X4 will remain uncomplemented. All four inputs will be put into a AND gate, resulting in $\overline{X1} X2 \overline{X3} X4$.

2. Product of Sum (POS)

Table used for **Z1**

No.	X1	X2	X3	X4	Z1	Result in Boolean Expression
<u>1</u>	0	0	0	0	1	-
<u>2</u>	0	0	0	1	0	$X1 + X2 + X3 + \overline{X4}$
<u>3</u>	0	0	1	0	1	-
<u>4</u>	0	0	1	1	0	$X1 + X2 + \overline{X3} + \overline{X4}$
<u>5</u>	0	1	0	0	0	$X1 + \overline{X2} + X3 + X4$
<u>6</u>	0	1	0	1	1	-
<u>7</u>	0	1	1	0	1	-
<u>8</u>	0	1	1	1	1	-
<u>9</u>	1	0	0	0	0	$\overline{X1} + X2 + X3 + X4$
<u>10</u>	1	0	0	1	1	-
<u>11</u>	1	0	1	0	1	-
<u>12</u>	1	0	1	1	1	-
<u>13</u>	1	1	0	0	0	$\overline{X1} + \overline{X2} + X3 + X4$
<u>14</u>	1	1	0	1	1	-
<u>15</u>	1	1	1	0	1	-
<u>16</u>	1	1	1	1	0	$\overline{X1} + \overline{X2} + \overline{X3} + \overline{X4}$

$$Z1 = (X1 + X2 + X3 + \overline{X4}) \times (X1 + X2 + \overline{X3} + \overline{X4}) \times (X1 + \overline{X2} + X3 + X4) \times (\overline{X1} + X2 + X3 + X4) \times (\overline{X1} + \overline{X2} + X3 + X4) \times (\overline{X1} + \overline{X2} + \overline{X3} + \overline{X4})$$

Table used for **Z2**

No.	X1	X2	X3	X4	Z2	Result in Boolean Expression
<u>1</u>	0	0	0	0	0	$X1 + X2 + X3 + X4$
<u>2</u>	0	0	0	1	1	-
<u>3</u>	0	0	1	0	0	$X1 + X2 + \overline{X3} + X4$
<u>4</u>	0	0	1	1	1	-
<u>5</u>	0	1	0	0	1	-
<u>6</u>	0	1	0	1	0	$X1 + \overline{X2} + X3 + \overline{X4}$
<u>7</u>	0	1	1	0	1	-
<u>8</u>	0	1	1	1	0	$X1 + \overline{X2} + \overline{X3} + \overline{X4}$
<u>9</u>	1	0	0	0	1	-
<u>10</u>	1	0	0	1	0	$\overline{X1} + X2 + X3 + \overline{X4}$
<u>11</u>	1	0	1	0	1	-
<u>12</u>	1	0	1	1	1	-
<u>13</u>	1	1	0	0	0	$\overline{X1} + \overline{X2} + X3 + X4$
<u>14</u>	1	1	0	1	0	$\overline{X1} + \overline{X2} + X3 + \overline{X4}$
<u>15</u>	1	1	1	0	1	-
<u>16</u>	1	1	1	1	1	-

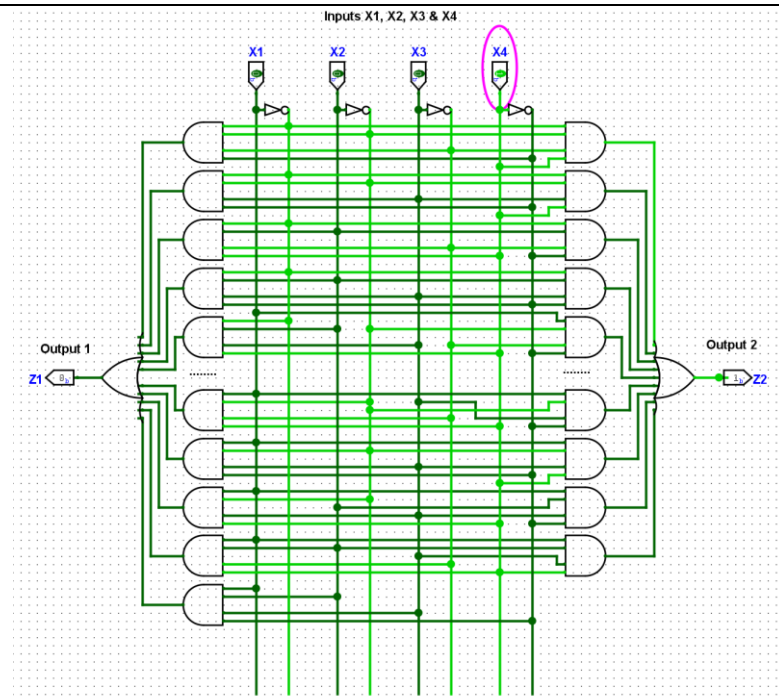
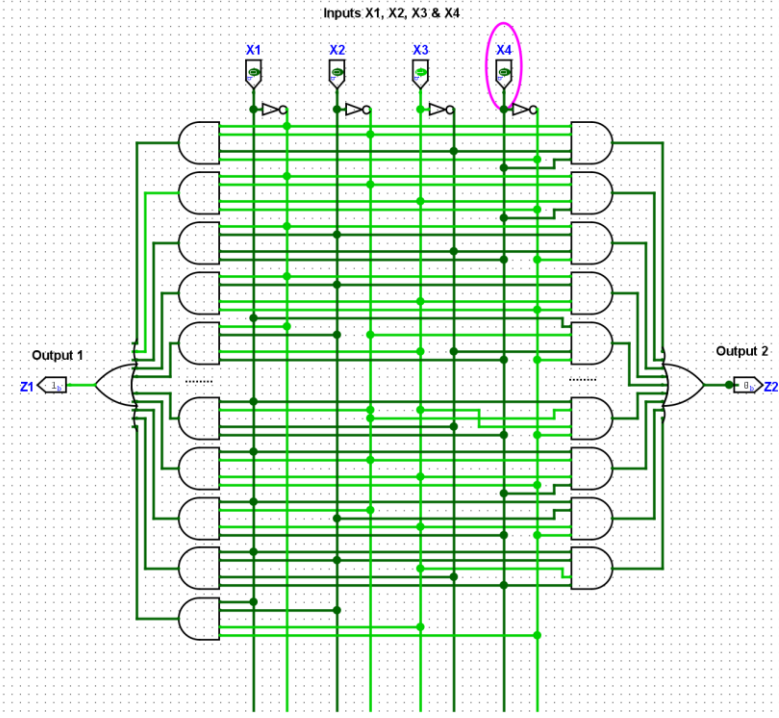
$$Z2 = (X1 + X2 + X3 + X4) \times (X1 + X2 + \overline{X3} + X4) \times (X1 + \overline{X2} + X3 + \overline{X4}) \times (X1 + \overline{X2} + \overline{X3} + \overline{X4}) \times (\overline{X1} + X2 + X3 + \overline{X4}) \times (\overline{X1} + \overline{X2} + X3 + X4) \times (\overline{X1} + \overline{X2} + X3 + \overline{X4})$$

Product of Sum (POS) Explanation:

- For each of the row that output equals to 0, we would find the **maxterm**.
- To obtain the maxterm, the inputs that give 0 remain uncomplemented, while the inputs that are 1 are complemented.
- Sum the variables together (put through an OR gate) to create one maxterm.
- Lastly, multiply (AND gate) all the maxterms.
- For example, in row 13 for Z2, X1 = 1, X2 = 1, X3 = 0, X4 = 0. X1 and X2 will be complemented, while X3 and X4 remain uncomplemented. Putting all through an OR gate, will result in the Boolean expression of $\overline{X1} + \overline{X2} + X3 + X4$.

Task 2.2: Logical circuit in Logisim

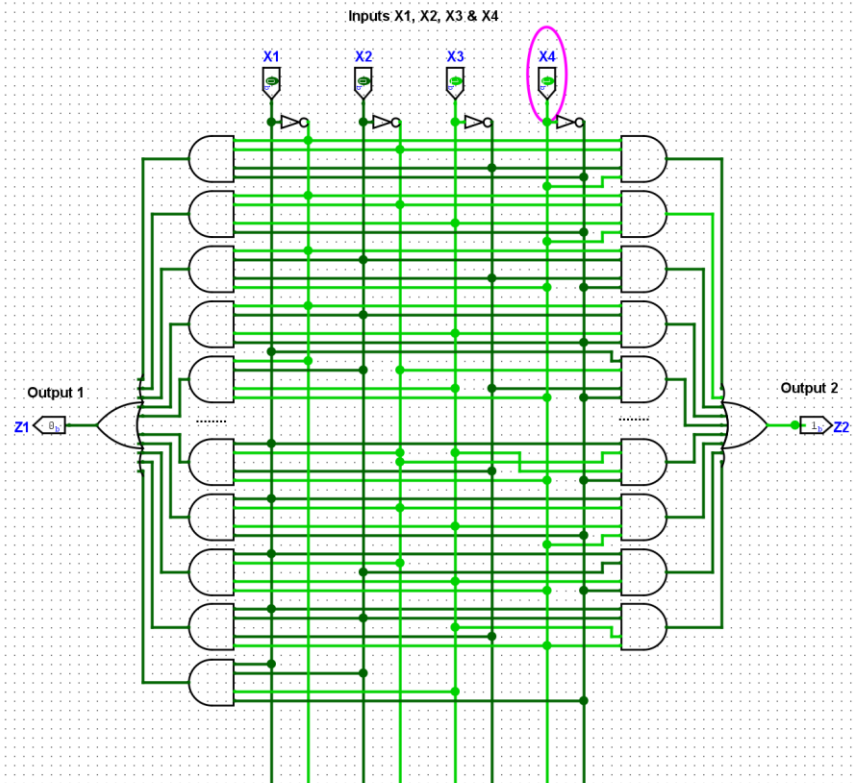
Test Cases for Z_1 and Z_2 :

Inputs	Output (Logisim)								
<table><tr><td>X1</td><td>X2</td><td>X3</td><td>X4</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr></table> Expected Output: • Z1 = 0 • Z2 = 1	X1	X2	X3	X4	0	0	0	1	
X1	X2	X3	X4						
0	0	0	1						
<table><tr><td>X1</td><td>X2</td><td>X3</td><td>X4</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr></table> Expected Output: • Z1 = 1 • Z2 = 0	X1	X2	X3	X4	0	0	1	0	
X1	X2	X3	X4						
0	0	1	0						

X1	X2	X3	X4
0	0	1	1

Expected Output:

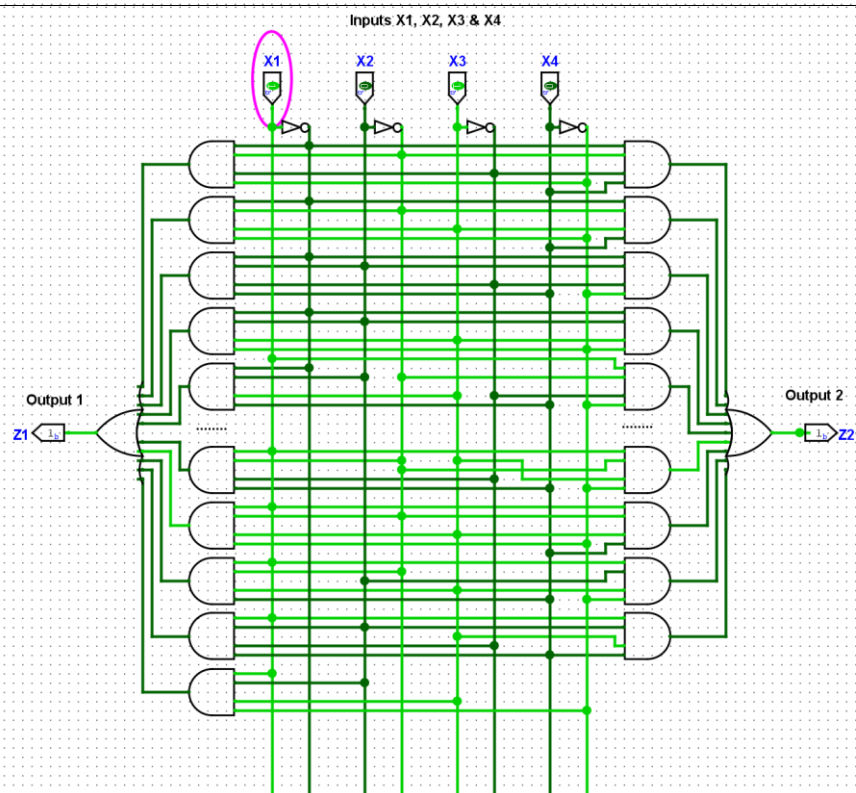
- Z1 = 0
- Z2 = 1



X1	X2	X3	X4
1	0	1	0

Expected Output:

- Z1 = 1
- Z2 = 1



Explanation:

- Firstly, the inputs for X1, X2, X3, and X4 are created.
- Both its uncomplemented and complemented input are made. The complemented input is made by passing it through a NOT gate.
- Each AND gate is made according to the Boolean function that was made earlier.
- Each AND gate is represented by one of the products minterm in the Boolean function, and the inputs are the respective variables in the minterm.
- Once all the AND gates are made, an OR gate is created with the inputs connected to the output of the AND gates.
- Lastly, the OR gate is connected to an output, and they are labelled accordingly.
- For example, Z1 circuits has 9 AND gates with 4 inputs, which are connected to one OR gate with 9 inputs

Task 2.3: Optimised circuit

General Explanation:

- This Karnaugh Map (K-Map) is from based on the values from the truth table.
- The rows represent values of X1 and X2, while the columns represent X3 and X4
- The header values for the row and columns are **2-bit grey codes**, where it's a sequence of binary numbers where each successive grey code only differ by one bit.
- The rest of the table is filled with their respective Z1/Z2 output, corresponding the combination of the row (X1 X2) and column (X3 X4).
- Once all 16 possible combinations have been filled in, grouping for Sum Of Products (SOP) can begin.
- The goal is to form **fewest** number of groups and **largest** possible groups.
- Each group represents one **AND** gate in the simplified Boolean formula. Where variables that change within the group are ignored, while the variables that stay the same form a term in the simplified formula (either X_i or $\overline{X_i}$).
- After all the AND gates are formed, the **OR** gate is used to string all the AND gates together, resulting in a simplified Boolean function.

K-Map for Z1:

$X3 X4$ $X1 X2$	00	01	11	10
00	1	0	0	1
01	0	1	1	1
11	0	1	0	1
10	0	1	1	1

$$Z1 = \overline{X1} \overline{X2} \overline{X4} + \overline{X1} X2 X4 + X1 \overline{X3} X4 + X1 \overline{X2} X4 + X3 \overline{X4}$$

Explanation:

- For $X3 \overline{X4}$, it is the largest group formed by grouping the green circle indicated on the K-Map. Since for all combinations, X3 remain as 1 and X4 remain as 0, X4 is passed through a NOT gate to become complemented so that the output is 1. X1 and X2 is ignored as they were both 0 and 1.
- For $\overline{X1} \overline{X2} \overline{X4}$, the cells are grouped using the wraparound rule, where the leftmost row is adjacent the right most row. Since variables X1, X2 and X3 all have the value of 0 for both outputs, they are all put complemented. X3 is ignored as it was both 0 and 1.
- $\overline{X1} X2 X4$ are grouped in the middle. Since X1 = 0, X2 = 1, X4 = 1 for both combinations, X1 is complemented. X3 is ignored as it was both 0 and 1.
- $X1 \overline{X3} X4$ are grouped. Since X1 = 1, X3 = 0, X4 = 1 for both combinations, X3 is complemented. X2 is ignored as it was both 0 and 1.
- $X1 \overline{X2} X4$ are the last group. For both combinations, X1 = 1, X2 = 0, X4 = 1, X3 is complemented. X3 is ignored as it was both 0 and 1.

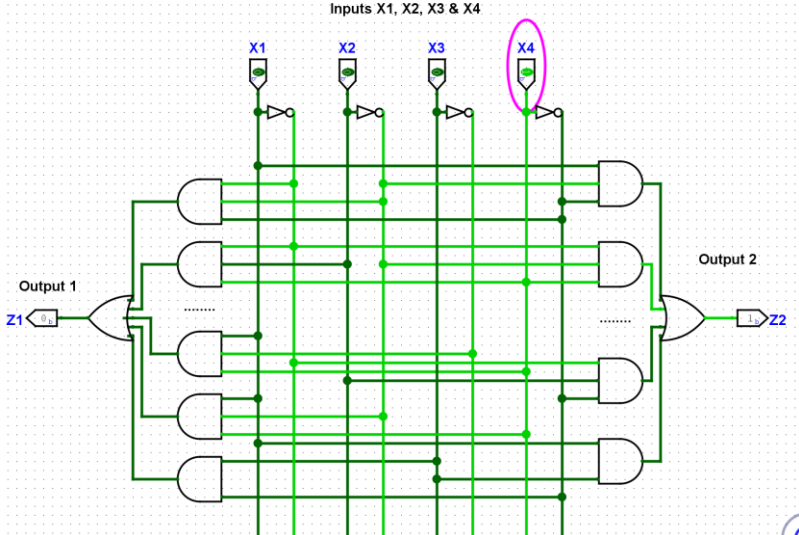
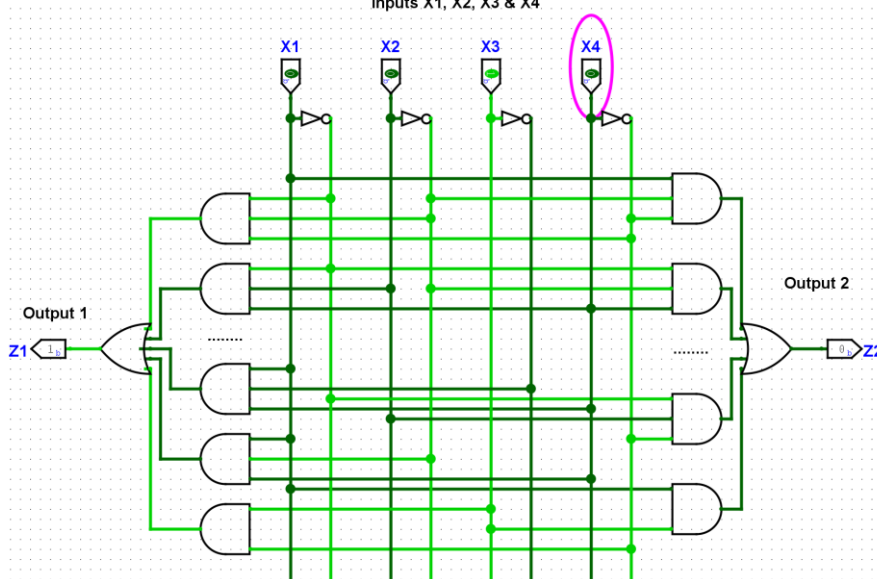
K-Map for Z2:

$X_3 X_4$ $X_1 X_2$	00	01	11	10
00	0	1	1	0
01	1	0	0	1
11	0	0	1	1
10	1		1	1

$$Z_1 = X_1 \overline{X_2} \overline{X_4} + \overline{X_1} \overline{X_2} X_4 + \overline{X_1} X_2 \overline{X_4} + X_1 X_3$$

Explanation:

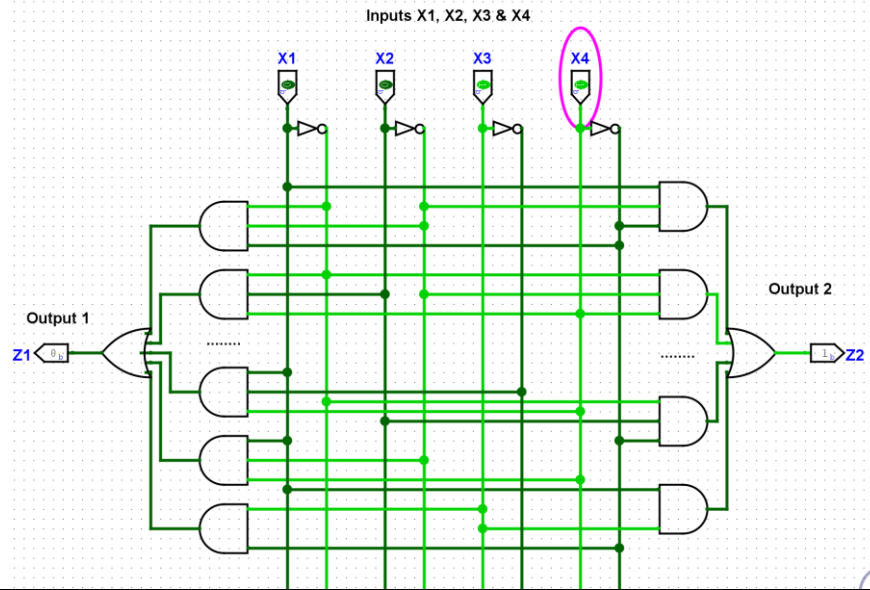
- For $X_1 X_3$, it is formed by the largest group possible. Out of the four combinations, X_1 remain as 1 and X_3 remain as 1 for all of them. Hence, $X_1 X_3$ is formed. X_2 and X_4 are ignored as they were both 1 and 0 for all the combinations.
- For $\overline{X_1} X_2 \overline{X_4}$, the group is formed by using the wraparound rule. Out of the two combinations, X_1 remain as 0, X_2 remain as 1 and X_4 remain as 0. Hence, $\overline{X_1} X_2 \overline{X_4}$ is formed.
- For $\overline{X_1} \overline{X_2} X_4$, the group is formed. Out of the two combinations, X_1 remain as 0, X_2 remain as 0 and X_4 remain as 1. Hence, $\overline{X_1} \overline{X_2} X_4$ is formed.
- For $X_1 \overline{X_2} \overline{X_4}$ the group is formed by using the wraparound rule. Out of the two combinations, X_1 remain as 1, X_2 remain as 0 and X_4 remain as 0. Hence, $X_1 \overline{X_2} \overline{X_4}$ is formed.

Inputs	Output (Logisim)								
<table><tr><th>X1</th><th>X2</th><th>X3</th><th>X4</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td></tr></table> Expected Output: • Z1 = 0 • Z2 = 1	X1	X2	X3	X4	0	0	0	1	Inputs X1, X2, X3 & X4 
X1	X2	X3	X4						
0	0	0	1						
<table><tr><th>X1</th><th>X2</th><th>X3</th><th>X4</th></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr></table> Expected Output: • Z1 = 1 • Z2 = 0	X1	X2	X3	X4	0	0	1	0	Inputs X1, X2, X3 & X4 
X1	X2	X3	X4						
0	0	1	0						

X1	X2	X3	X4
0	0	1	1

Expected Output:

- $Z1 = 0$
- $Z2 = 1$



X1	X2	X3	X4
1	0	1	0

Expected Output:

- $Z1 = 1$
- $Z2 = 1$

